Pump: Operation and Maintenance Manual

Applies to: Grundfos NK 80, KSB Etanorm 50, Wilo IL 100. Document no. OM-PUMP-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
pressure	outlet	bar	approx. 4	> 5.2	> 6
flow	outlet	l/min	approx. 320	> 420	> 470
vibration	bearing	mm/s	approx. 1.8	> 4	> 6.5
temperature	motor winding	degC	approx. 60	> 80	> 95

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High pressure (outlet, bar)

Probable causes:

- Discharge valve partially closed
- Clogged downstream filter
- Blocked or scaled discharge pipe

Corrective actions:

1. Check all discharge valve positions against the P&ID.
2. Measure differential pressure across the downstream filter; replace the element SP-PMP-FLT-80 if it exceeds 0.8 bar.
3. Never operate the pump against a closed valve for more than 30 seconds.

4.2 High flow (outlet, l/min)

Probable causes:

- Bypass valve open
- Leak or burst pipe downstream
- Pump operating at run-out (far right of the curve)

Corrective actions:

1. Verify the bypass valve is closed during normal operation.
2. Walk down the discharge line for leaks.
3. Throttle the discharge valve back to the best-efficiency point (320 l/min); run-out operation causes cavitation and motor overload.

4.3 High vibration (bearing, mm/s)

Probable causes:

- Cavitation due to low suction pressure (insufficient NPSH)
- Pump/motor misalignment
- Worn pump bearings or damaged impeller

Corrective actions:

1. Check suction pressure: it must stay above 0.5 bar. A gravel-like noise indicates cavitation.
2. Perform laser alignment (max offset 0.05 mm).
3. Replace the bearing kit SP-PMP-BRG-62 and inspect the impeller for erosion.

4.4 High temperature (motor winding, degC)

Probable causes:

- Motor overload
- Blocked motor cooling fan or fan cover
- Phase imbalance above 2%

Corrective actions:

1. Compare motor current to the nameplate rating.
2. Clean the fan cover; the motor needs 10 cm of free space for airflow.
3. Measure phase voltages; imbalance above 2% must be corrected by the electrical team.

5. Maintenance schedule

Interval	Task
Weekly	Check mechanical seal for leakage (max 10 drops/min)
Every 30 days	Check alignment and bearing temperature
Every 3 months	Replace downstream filter element SP-PMP-FLT-80
Every 12 months	Replace mechanical seal SP-PMP-SEAL-80